Chapter 2
Environmental Chemistry

Lu Elfa Peng

elfapeng@connect.hku.hk

2857 8470

Expected learning outcomes for Chapter 2

- **Stoichiometry**
- Oxygen demand
- **Chemical equilibrium**
  - **Reactor quotient (Q) and equilibrium constant (K)**
  - **P notation: pX = -log10(X)**
  - **Chemical speciation**
  - **Solubility**
  - **Henry' law**
- ~~Nuclear chemistry~~
  - ~~Radioactive decay and half life~~

Stoichiometry

- Balancing chemical reactions

  $CH_4 + O_2 \rightarrow CO_2 + H_2O$

- Atomic weight
  - C = 12 g/mol; H = 1 g/mol; O = 16 g/mol
- Molecular weight
  - CH_4: 12 + 1 x 4 = 16 g/mol; O_2:32 g/mol; CO_2 = 44 g/mol; H_2O = 18 g/mol

CH_4	+	$2O_2 \rightarrow$	CO_2	+	$2H_2O$
1 mol		2 mol	1 mol		2 mol
16 g		64 g	44 g		36 g

Stoichiometry

Ex. 2.2 Theoretical oxygen demand

A wastewater contains 1.67 **mM** glucose ($C_6H_{12}O_6$). Find the amount of oxygen required to fully oxidize glucose.

Solution:

$C_6H_{12}O_6$ +	$6O_2$ →	$6CO_2$ +	$6H_2O$
1	6		
1.67 mM	1.67 x 6 = 10 mM		

Theoretical oxygen demand = 10 x 10^{-3} mol/l x 32 g/mol = 320 x 10^{-3} g/l

= **320 mg/l**

Or simply, 10 mM x 32 mg/mmol = 320 mg/l

Oxygen demand

- amount of oxygen required to oxidize organic matter

- Theoretical oxygen demand
  - Calculated based on stoichiometry
  - Requires the knowledge of molecular formula
- Biochemical oxygen demand (BOD)
  - Oxygen required to biologically remove a (organic) pollutant
- Chemical oxygen demand (COD)
  - Oxygen required to chemically oxidize a (organic) pollutant
- Theoretical oxygen demand > COD > BOD

Chemical equilibria

Limitations
Reactions are not always at equilibrium
Some slow reactions

A general chemical reaction

$$aA + bB \rightleftharpoons cC + dD$$

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$ Q

Notes:

1. *K* is the equilibrium constant (largely dependent on temperature)
   - ❑ *Q* = *K*, reaction at equilibrium
   - ❑ *Q* < *K*, reaction proceeds forward →
   - ❑ *Q* > *K*, reaction proceeds backward ←
2. Concentration in M. Pure solid or pure liquid, activity = 1.

Acidity constant: K_a Solubility product: K_{sp} or K_s Henry's constant: K_H

Chemical equilibria

A general chemical reaction

$$aA + bB \rightleftharpoons cC + dD$$

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$ Q

Notes:

Pure liquid or pure solid, activity = 1.

Examples (just for your reference how the Kw for water and the Ksp for different solids come out by using the above generic equation):

Pure water: $H_2O \rightleftharpoons H^+ + OH^-$ $\quad K = \frac{[H^+]^1[OH^-]^1}{[H_2O]^1}$ →

$K*[H_2O] = \frac{[H^+]^1[OH^-]^1}{1} = Kw = [H^+][OH^-] = 10^{-14}$

Pure solids: $Solid(s) \rightleftharpoons aA + bB$ $\quad K = \frac{[A]^a[B]^b}{[Solid]^1}$ →

$K*[Solid] = \frac{[A]^a[B]^b}{1} = Ksp = [A]^a[B]^b$

Kw and Ksp can be obtained in handbooks

Chemical equilibria

- acid-base equilibria

$$HA \rightleftharpoons H^+ + A^- \qquad K_a = \frac{[H^+][A^-]}{[HA]}$$

p notation

$pH = -\log_{10}[H^+]$

$pC = -\log_{10}C$

$pK_a = -\log_{10}K_a$

…

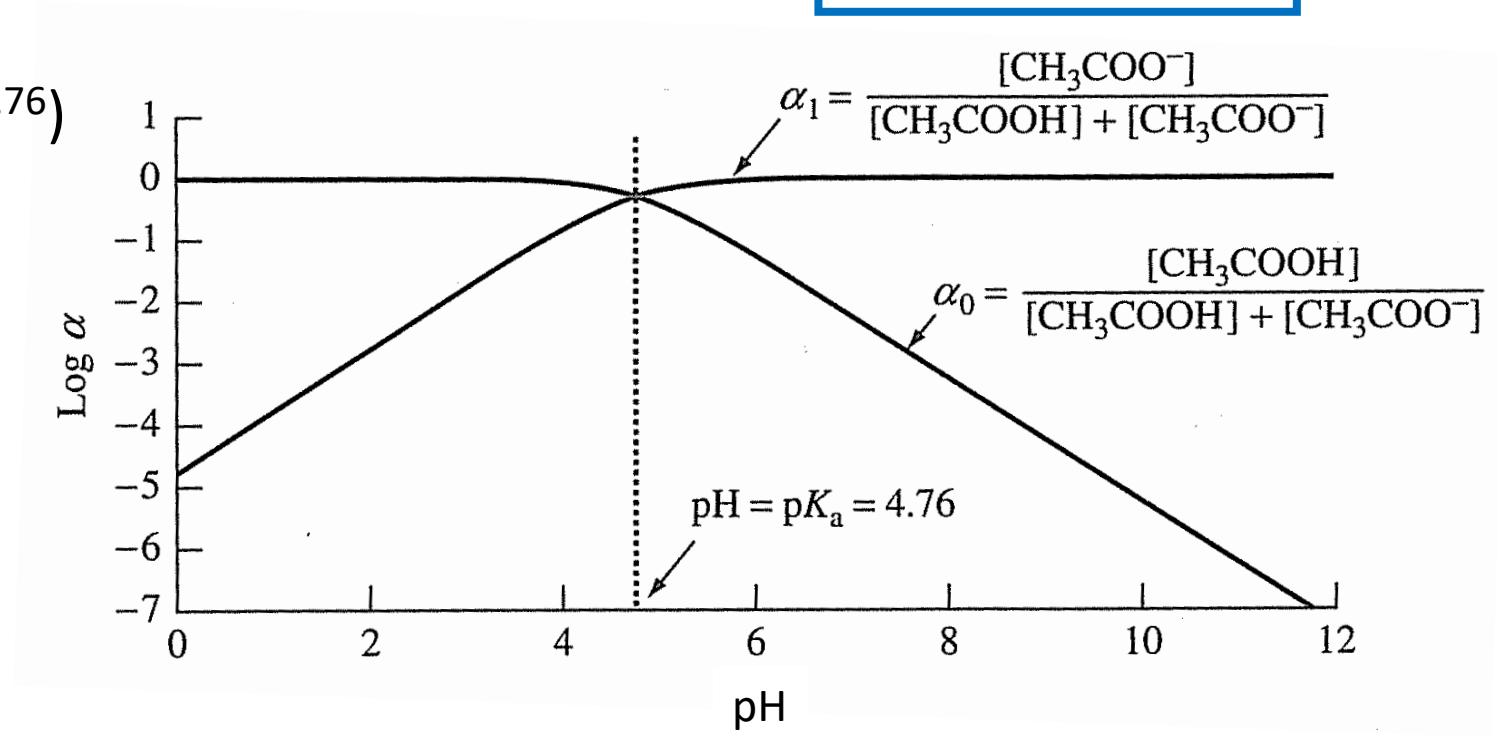
$pX = -\log_{10}X$

For acetic acid (CH_3COOH or HAc),

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ $(K_a = 10^{-4.76})$

$$K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$$

➔ $$\frac{[CH_3COO^-]}{[CH_3COOH]} = \frac{K_a}{[H^+]}$$



Chemical equilibria

- acid-base equilibria

$$\underset{\text{acid}}{HA} \rightleftharpoons \underset{\text{proton}}{H^+} + \underset{\text{base}}{A^-}$$ (Lewis acids and bases) $$K_a = \frac{[H^+][A^-]}{[HA]}$$

p notation

$pH = -\log_{10}[H^+]$

$pC = -\log_{10}C$

$pK_a = -\log_{10}K_a$

...

$pX = -\log_{10}X$

A weak acid (HA) in water: part of HA will become A-: $[HA] \rightleftharpoons [A^-]$

Add a certain mass of HA in a cup of water → with an initial concentration of $[HA]_0$

$[HA]_0$ = Ct (total concentration) = $[HA]+[A^-]$

$$\frac{[A^-]}{[HA]} = \frac{K_a}{[H^+]} \quad ①$$

Ka is a constant and commonly you will know pH and Ct, so you can use these two equations to solve the two unknowns of [HA] and $[A^-]$.

$Ct = [HA]+[A^-]$ ②

Chemical equilibria - acid-base equilibria

! ! ! This is an equilibrium expression, not a concentration equation.

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$HA \rightleftharpoons A^- + H^+$

($K_a = 10^{-4.76}$)

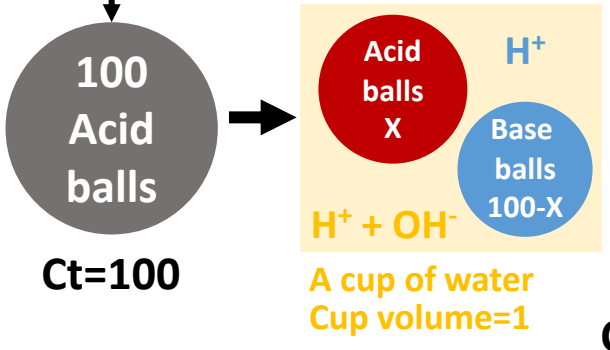
$$HA \rightleftharpoons H^+ + A^- \qquad K_a = \frac{[H^+][A^-]}{[HA]}$$

Example:

Part of these acid balls will stay as **acid balls**, but another part will become **base balls** and H^+ after putting into the **water**

This is a concentration equation.

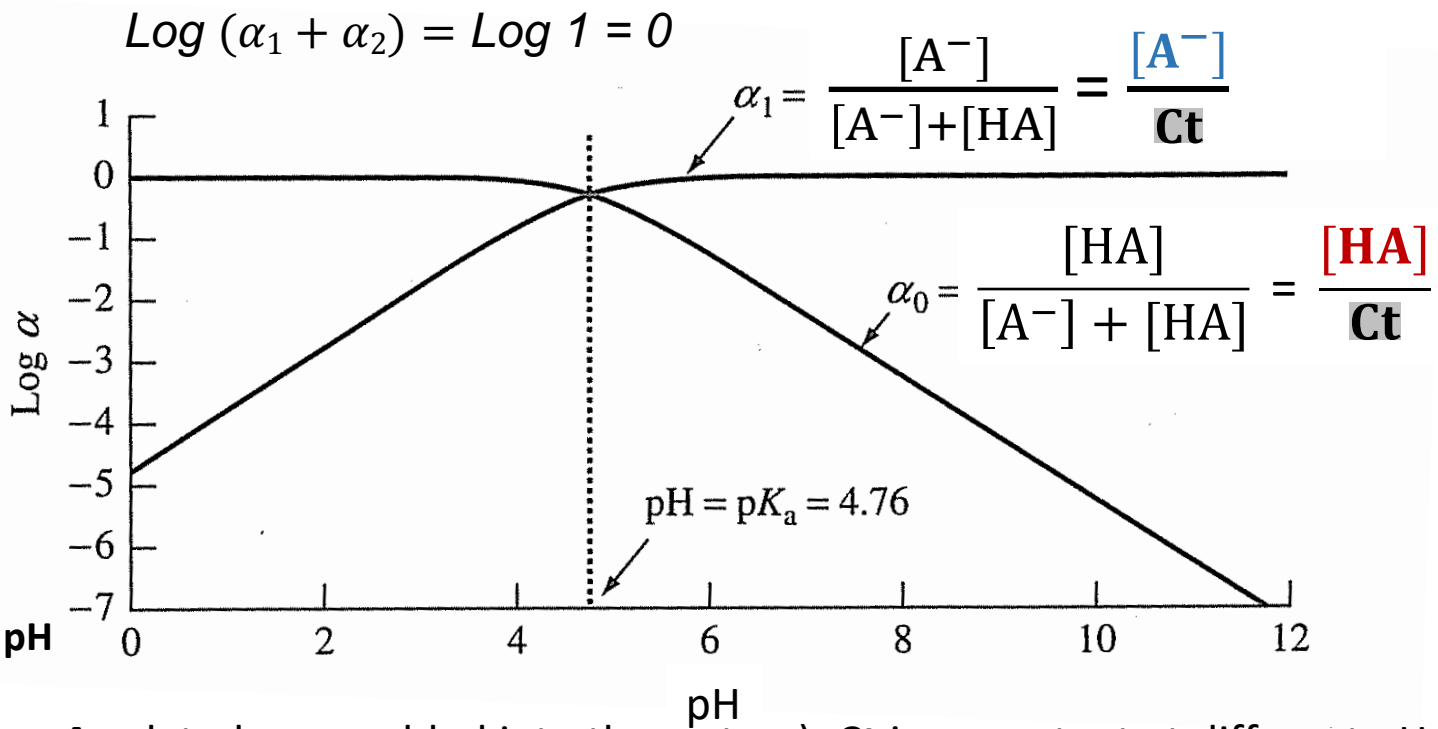
Ct = [HA] + [A^-]



[Red balls] = X

[Blue balls] = (100-X)

[H^+ + H^+] indicated by **pH**



Close system (no more **A**-related mass added into the system): **Ct** is a constant at different pH

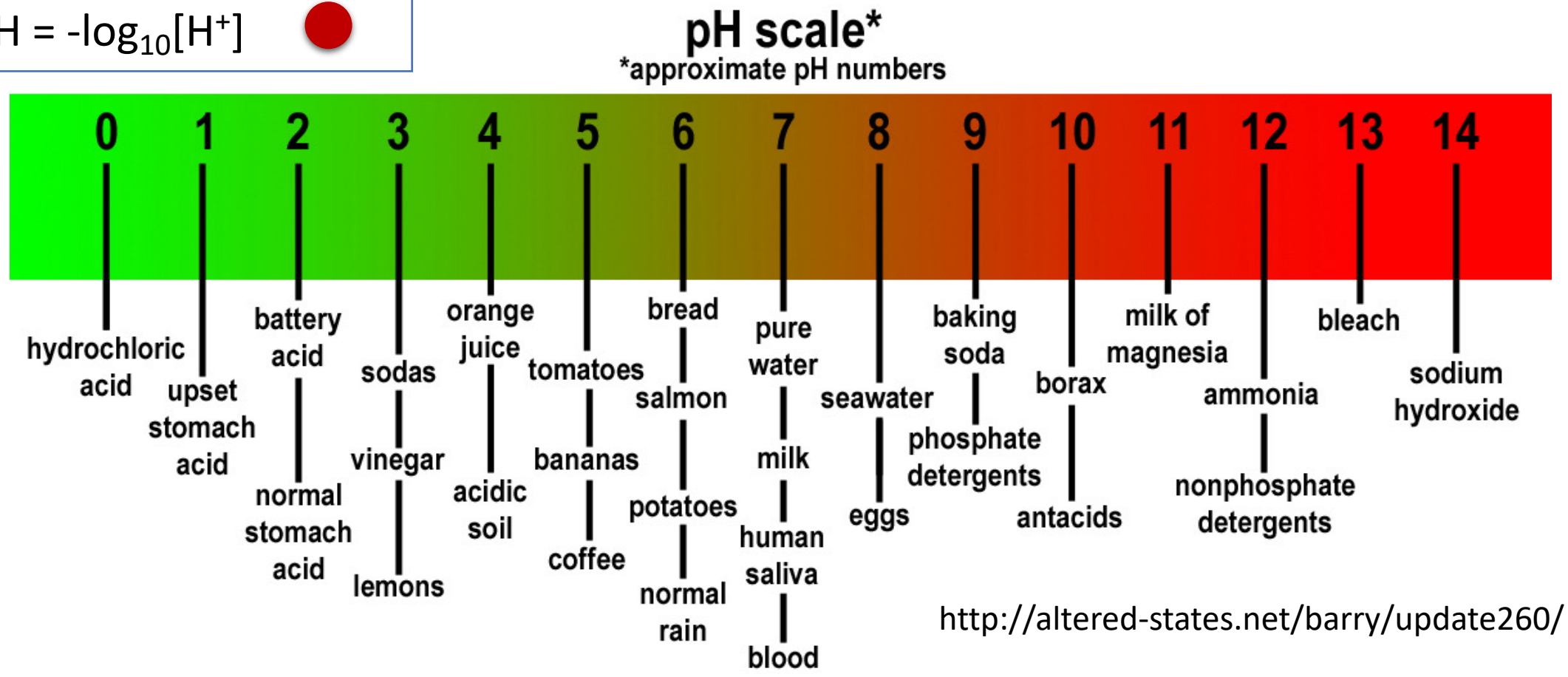
Chemical equilibria

- pH scale

$[OH^-] = Kw/[H^+] = 10^{-14} /[H^+]$

p notation

$pH = -\log_{10}[H^+]$



http://altered-states.net/barry/update260/

Chemical equilibria

- acid-base equilibria

p notation

pH = $-\log_{10}[H^+]$

pC = $-\log_{10}C$

$pK_a = -\log_{10}K_a$

...

pX= $-\log_{10}X$

Table 6.2

Dissociation Constants and Chemical Formulas for Some Acids of Environmental Importance

Acid	Formula	pK_{a1}	pK_{a2}	pK_{a3}
Nitric acid	HNO_3	−1.30	—	—
Hydrochloric acid	HCl	<0	—	—
Sulfuric acid	H_2SO_4	<0	1.99	—
Phosphoric acid	H_3PO_4	2.16	7.20	12.35
Arsenic acid	H_3AsO_4	2.24	6.76	11.5
Citric acid	$C_3H_4OH(COOH)_3$	3.13	4.72	6.33
Acetic acid	CH_3COOH	4.76	—	—
Carbonic acid	H_2CO_3	6.35	10.33	—
Hydrogen sulfide	H_2S	6.99	12.92	—
Hypochlorous acid	HOCl	7.60	—	—
Ammonium ion	NH_4^+	9.26	—	—
Silicic acid	H_4SiO_4	9.84	13.20	—

Chemical equilibria

- acid-base equilibria

Ex. 2.8 Disinfection

Hypochlorous acid (HOCl) is a much stronger disinfectant than hypochlorite (OCl^-).

$$HOCl \rightleftharpoons OCl^- + H^+$$

The equilibrium constant equation is

$$K_{a1} = \frac{[OCl^-][H^+]}{[HOCl]} = 10^{-7.60}$$

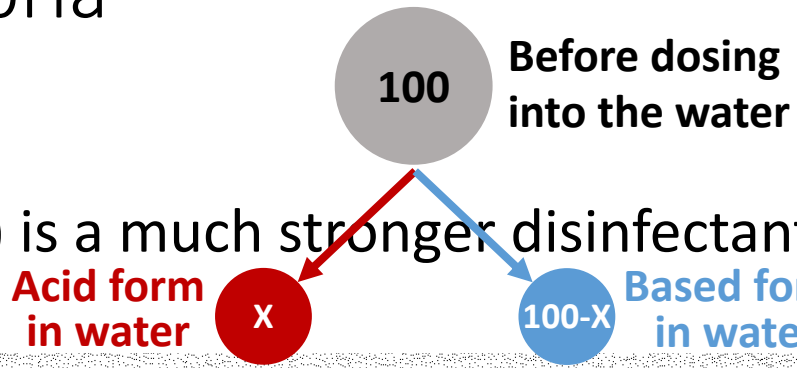
What is the minimum pH required to make sure 99% of the chemical exists in the acid form?

Chemical equilibria

- acid-base equilibria

Ex. 2.8 Disinfection

Hypochlorous acid (HOCl) is a much stronger disinfectant than hypochlorite (OCl^-).



$$HOCl \rightleftharpoons OCl^- + H^+$$

The equilibrium constant equation is

$$K_{a1} = \frac{[OCl^-][H^+]}{[HOCl]} = 10^{-7.60}$$

$$99\% = \frac{99}{100} \qquad 1\% = \frac{1}{100}$$

What is the minimum pH required to make sure **99%** of the chemical exists in the **acid form**?

$$\frac{1}{99} = \frac{K_{a1}}{[H^+]} \rightarrow pH = 5.6$$

notes

Given $NH_4^+ \rightleftharpoons NH_3 + H^+$ with the constant of pKa = 9.3, what is the acid in the equilibrium, and what is the ratio of $[NH_3]$ over $[NH_4^+]$ at pH 10.3, respectively?

(A) NH_4^+, 10; (B) NH_4^+, 0.1; (C) NH_3, 10; (D) NH_3, 0.1.

Chemical equilibria
- Solubility product

K_{sp}: the equilibrium constant for dissolving a sparingly soluble ionic solid into its ions in a saturated solution given a certain temperature.

$$\text{Solid (s)} \rightleftharpoons aA + bB \qquad K_{sp} = [A]^a[B]^b$$

1g $CaSO_4$(s) is added into 1 L of pure water. Immediately, the solid begins to dissolve following the reaction: $CaSO_4(s) \rightleftharpoons Ca^{2+}(aq) + SO_4^{2-}(aq)$. What is the concentration of dissolved sulfate in water at equilibrium ($K_{sp,\ CaSO4} = 2*10^{-5}$)?

$CaSO_4(s) \rightleftharpoons$	$Ca^{2+}(aq)$ +	$SO_4^{2-}(aq)$
Dosing x	x	x
$(1\ g/136\ g/mol)/1L$ = 7.3 mM	7.3 mM	7.3 mM

$K_{sp,\ CaSO4} = [Ca^{2+}]^1[SO_4^{2-}]^1 = 2*10^{-5}$

$Q = (7.3*10^{-3})^2 = 5.3*10^{-5} > K_{sp} = 2*10^{-5}$

Oversaturated

$[SO_4^{2-}] = \sqrt{K_{sp,\ CaSO4}} = \sqrt{2*10^{-5}} = 4.5$ mM

Chemical equilibria

- Solubility product

$\text{Solid (s)} \rightleftharpoons aA + bB \qquad K_{sp} = [A]^a[B]^b$

Ex. 2.9 Fluoride solubility in saturated solution

CaF_2 (s) $\rightleftharpoons Ca^{2+} + 2F^-$ ($K_{sp} = 3 \times 10^{-11}$)

c 2c

$[Ca^{2+}][F^-]^2 = 3 \times 10^{-11}$

$c\ (2c)^2 = 3 \times 10^{-11}$

$c = 2 \times 10^{-4}$ M

$[F^-] = 4 \times 10^{-4}$ M or 15.2 mg/L

Chemical equilibria

- Solubility product

$\text{Solid (s)} \rightleftharpoons aA + bB \qquad K_{sp} = [A]^a[B]^b$

Fluoride solubility in groundwater (initially contains 3 mM Ca^{2+})

2×10^{-4}

CaF_2 (s) $\rightleftharpoons Ca^{2+} + 2F^-$ ($K_{sp} = 3 \times 10^{-11}$)

Rough calculation
(ignore the Ca^{2+} from CaF_2):
$[Ca^{2+}][F^-]^2 = 3 \times 10^{-11}$
$3 \times 10^{-3} * [F^-]^2 = 3 \times 10^{-11}$
[F-] = **10^{-4} M**

For accurate calculation	Ca^{2+}	F^-
Initial concentration in water	3×10^{-3}	0
Concentration contributed by CaF2	c	2c
Final concentration	$3 \times 10^{-3} + c$	2c

Chemical equilibria

- Gas liquid partitioning (solubility of gas)

Where
[X] is the concentration of dissolved gas;
P_x is the partial pressure of the gas

- Henry's Law

X in the air ⇌ X dissolved in the water

$$[\mathrm{X}] = K_H P_x$$

Chemical equilibria

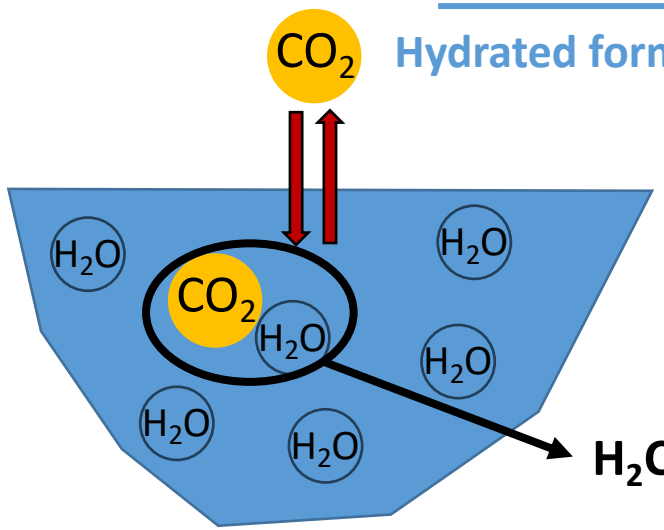
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Where
[X] is the concentration of dissolved gas;
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- Henry's Law

X in the air ⇌ X dissolved in the water $[X] = K_H P_x$

$CO_2(g) + H_2O \rightleftharpoons H_2CO_3(aq)$ $[H_2CO_3] = K_H P_{CO2}$ ($K_H = 0.0334 = 10^{-1.5}$ mol/L.atm)



Hydrated form of CO_2 in water

Assume that the concentration of CO_2 is 390 ppmv, what is $[H_2CO_3]$?

Concentration of dissolved CO_2 in water
$= [H_2CO_3] = K_H P_{CO_2}$
$= 0.0334 \text{ mol/(L·atm)} \times 390 \times 10^{-6} \text{ atm} = 1.3 \times 10^{-5} \text{ mol/L}$

$H_2CO_3(aq)$ (When CO_2 is dissolved in water, it exists in the form of H_2CO_3)

The ocean is an **open system** that is open to the atmosphere,
where Ct of the carbonate species in the ocean is not a constant when pH changes

notes

Chemical equilibria

- Gas liquid partitioning (solubility of gas)

Where
[X] is the concentration of dissolved gas;
P_x is the partial pressure of the gas

- Henry's Law

X in the air ⇌ X dissolved in the water $[X] = K_H P_x$

Ex. 2.10 Oxygen solubility

K_H = 0.00126 mol/L.atm at 25 °C

At ground level, partial pressure of oxygen is 0.21 atm

$[O_2]$ = 0.00126 x 0.21 = 2.65 x 10^{-4} mol/l → **8.5 mg/l**

Atmospheric pressure decreases at higher elevation:

$P = P_0 - 1.15 \times 10^{-4} H$ (*H* indicates height)

At higher elevation, solubility of oxygen will be significantly decreased.

Read Ex. 2.11

Chemical equilibria

Limitations
Reactions are not always at equilibrium
Some slow reactions

A general chemical reaction

$$aA + bB \rightleftharpoons cC + dD$$

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Q

Henry's constant: K_H
gas-liquid equilibrium (two-phase)

$CO_2(g) + H_2O \rightleftharpoons H_2CO_3(aq)$ $[H_2CO_3] = K_H P_{CO2}$

Acidity constant: K_a
acid-base equilibrium (one-phase)

$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ $K_{a1} = [H^+][HCO_3^-]/[H_2CO_3]$

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$ $K_{a2} = [H^+][CO_3^{2-}]/[HCO_3^-]$

Solubility product: K_{sp} or K_s
solid-liquid equilibrium (two-phase)

$CaCO_3\ (s) \rightleftharpoons Ca^{2+} + CO_3^{2-}$ $K_{sp} = [Ca^{2+}][CO_3^{2-}]$

notes

Chemical equilibria

- Carbonate system

Air-water equilibrium

$CO_2\ (g) + H_2O \rightleftharpoons H_2CO_3\ (aq)$ $\quad [H_2CO_3] = K_H P_{CO2}$ $\quad (K_H = 0.0334 = 10^{-1.5}$ mol/L.atm)

Acid-base equilibrium

$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ $\quad K_1 = [H^+][HCO_3^-]/[H_2CO_3] = 4.47 \times 10^{-7} = 10^{-6.35}$ $\quad (pK_1 = 6.35)$

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$ $\quad K_2 = [H^+][CO_3^{2-}]/[HCO_3^-] = 4.68 \times 10^{-11} = 10^{-10.33}$ $\quad (pK_2 = 10.33)$

Liquid-solid equilibrium

$CaCO_3\ (s) \rightleftharpoons Ca^{2+} + CO_3^{2-}$ $\quad K_{sp} = [Ca^{2+}][CO_3^{2-}] = 4.57 \times 10^{-9} = 10^{-8.3}$ $\quad (pK_{sp} = 8.3)$

Chemical equilibria

- acid-base equilibria

$$HA \rightleftharpoons H^+ + A^- \quad K_a = \frac{[H^+][A^-]}{[HA]}$$

$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$

$K_{a1} = [H^+][HCO_3^-]/[H_2CO_3] = 4.47 \times 10^{-7}$

($pK_{a1} = 6.35$)

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$

$K_{a2} = [H^+][CO_3^{2-}]/[HCO_3^-] = 4.68 \times 10^{-11}$

($pK_{a2} = 10.33$)

$Ct = [H_2CO_3] + [HCO_3^-] + [CO_3^{2-}]$

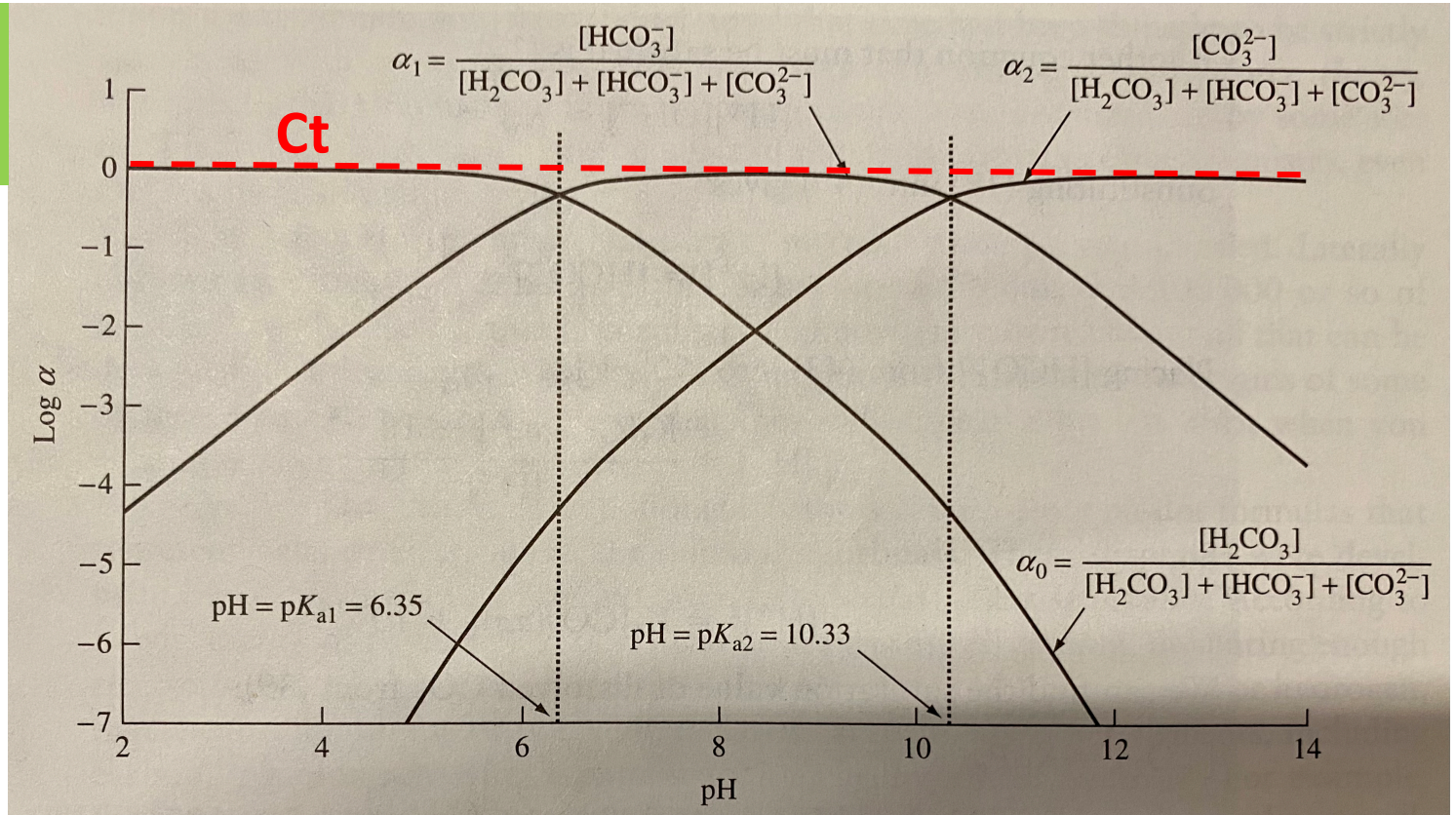


FIGURE 5 α diagram for the carbonate system. α_0 is the fraction of the total carbonate in the system that is in the carbonic acid form (H_2CO_3), α_1 is the fraction in the bicarbonate anion form (HCO_3^-), and α_2 is the fraction in the carbonate anion form (CO_3^{2-}).

Chemical equilibria

- Carbonate system

Example:

Know $Pco_2=10^{-3.5}$ atm and $K_H = 10^{-1.5}$ M/atm, what are $[HCO_3^-]$ and $[CO_3^{2-}]$ at pH7.3?

$[H_2CO_3]$ = $Pco_2 \times K_H = 10^{-5}$ M

$[HCO_3^-]/[H_2CO_3]=K_1/[H^+]$ = $10^{-6.3}/10^{-7.3} = 10$ (K1 = $10^{-6.3}$ and **$[H^+] = 10^{-7.3}$**)

→ **$[HCO_3^-]$** = $[H_2CO_3] \times 10 = 10^{-5} \times 10 = 10^{-4}$ M (at pH7.3)

$[CO_3^{2-}]/[HCO_3^-] = K_2/[H^+]$ = $10^{-10.3}/10^{-7.3} = 10^{-3}$

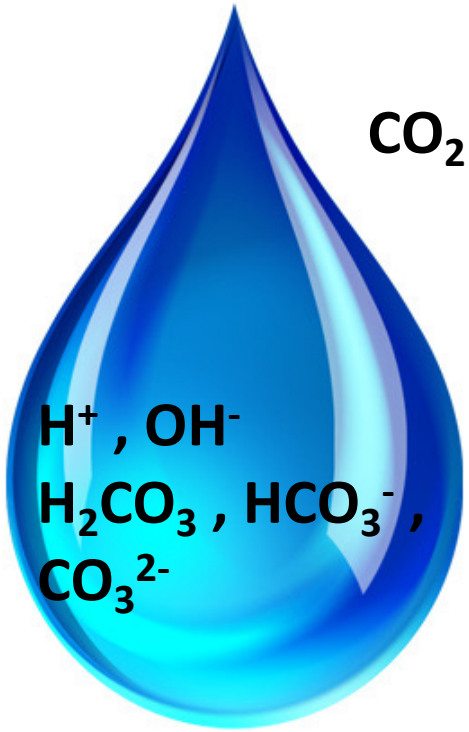
→ **$[CO_3^{2-}]$** = $[HCO_3^-] \times 10^{-3} = 10^{-4} \times 10^{-3} = 10^{-7}$ M (at pH7.3)

At this pH, this ratio is low, $[CO_3^{2-}] << [HCO_3^-]$, so $[CO_3^{2-}]$ can be ignored in some cases

Chemical equilibria

- pH of rain

$pK_1 = 6.3$, $pK_2 = 10.3$, $Kw=10^{-14}$



CO_2 = 390 ppm

Air-water equilibrium

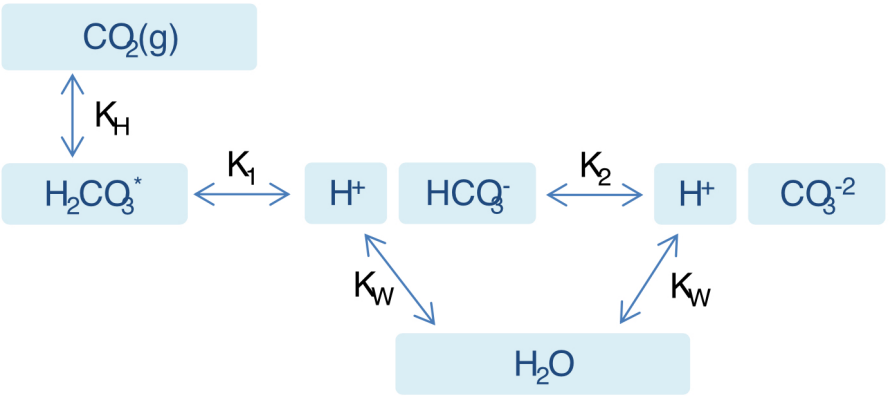
$[H_2CO_3] = K_H P_{CO2} = 0.0334 \times 390 \times 10^{-6} = 1.3 \times 10^{-5}$ M

Acid-base equilibrium

$[HCO_3^-] = [H_2CO_3] \times (K_1/[H^+])$

$[CO_3^{2-}] = [HCO_3^-] \times (K_2/[H^+])$

$[OH^-] = K_w/[H^+] = 10^{-14}/[H^+]$



Charge balance

$[H^+] = [HCO_3^-] + 2[CO_3^{2-}] + [OH^-]$

$[H^+] = [H_2CO_3] \times (K_1/[H^+]) + 2([HCO_3^-] \times (K_2/[H^+])) + 10^{-14}/[H^+]$

$[H^+] = 1.3 \times 10^{-5} \times 10^{-6.3}/[H^+] + 2((1.3 \times 10^{-5} \times 10^{-6.3}/[H^+]) \times (10^{-6.3}/[H^+])) + 10^{-14}/[H^+]$

👆 So complicated calculation for $[H^+]$, you may solve it by yourself if interested.

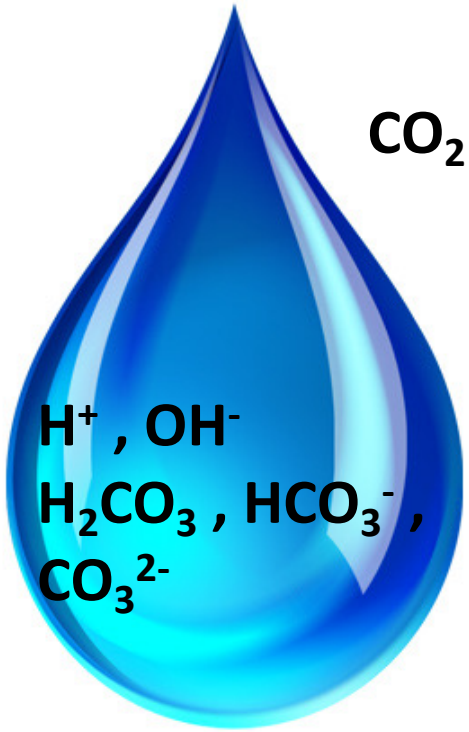
example

Chemical equilibria

- pH of rain

Water + acid → pH < 7,
Assume pH=6, $[H^+] = 10^{-6}$

$pK_1 = 6.3$, $pK_2 = 10.3$, $Kw=10^{-14}$



CO_2 = 390 ppm

Air-water equilibrium

$[H_2CO_3] = K_H P_{CO2} = 0.0334 \times 390 \times 10^{-6} = 1.3 \times 10^{-5}$ M

Acid-base equilibrium

$[HCO_3^-] = [H_2CO_3] \times (K_1/[H^+])$

$[CO_3^{2-}] = [HCO_3^-] \times (K_2/[H^+])$

$[OH^-] = K_w/[H^+] = 10^{-14}/[H^+]$

Order-of-Magnitude Estimation

$[HCO_3^-] = 10^{-5} * 10^{-6.3}/10^{-6} = 10^{-5.3}$

$[CO_3^{2-}] = 10^{-5.3} * 10^{-10.3}/10^{-6} = 10^{-9.6}$

$[OH^-] = 10^{-14}/10^{-6} = 10^{-8}$

Charge balance

$[H^+] = [HCO_3^-] + 2[CO_3^{2-}] + [OH^-]$

$[H^+] = [H_2CO_3] \times (K_1/[H^+])$

$[H^+] = (K_1[H_2CO_3])^{0.5} = (4.47 \times 10^{-7} \times 1.3 \times 10^{-5})^{0.5} = 2.42 \times 10^{-6}$ M

pH = 5.62

Chemical equilibria

- Acid rain

Example:

Knowns: CO_2 = 390 ppm in the air and **1mM HNO_3** in the rainwater,

Question: what's the rainwater pH?

$HNO_3 \rightarrow NO_3^- + H^+$ (Strong acid)

1mM **1mM → $[H^+] = 0.001M = 10^{-3}$ → pH = 3**

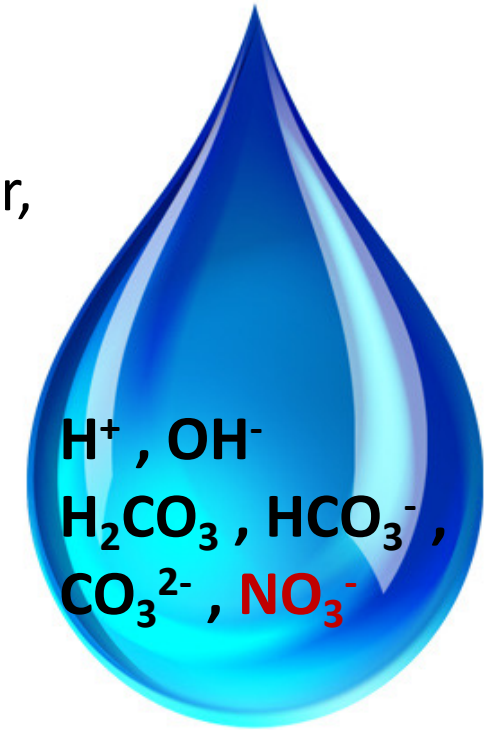
Order-of-Magnitude Estimation by assuming pH = 3

$[HCO_3^-] = 10^{-5} * 10^{-6.3} / 10^{-3} = 10^{-8.3}$

$[CO_3^{2-}] = 10^{-8.3} * 10^{-10.3} / 10^{-3} = 10^{-15.6}$

$[OH^-] = 10^{-14} / 10^{-3} = 10^{-11}$

All these species can be ignored, so pH = 3.



Air pollutants: SO_2 (later oxidized to SO_3), NO_x . They will dissolve into water to be strong acids.

$SO_3 + H_2O \rightarrow H_2SO_4 \rightarrow SO_4^{2-} + 2H^+$

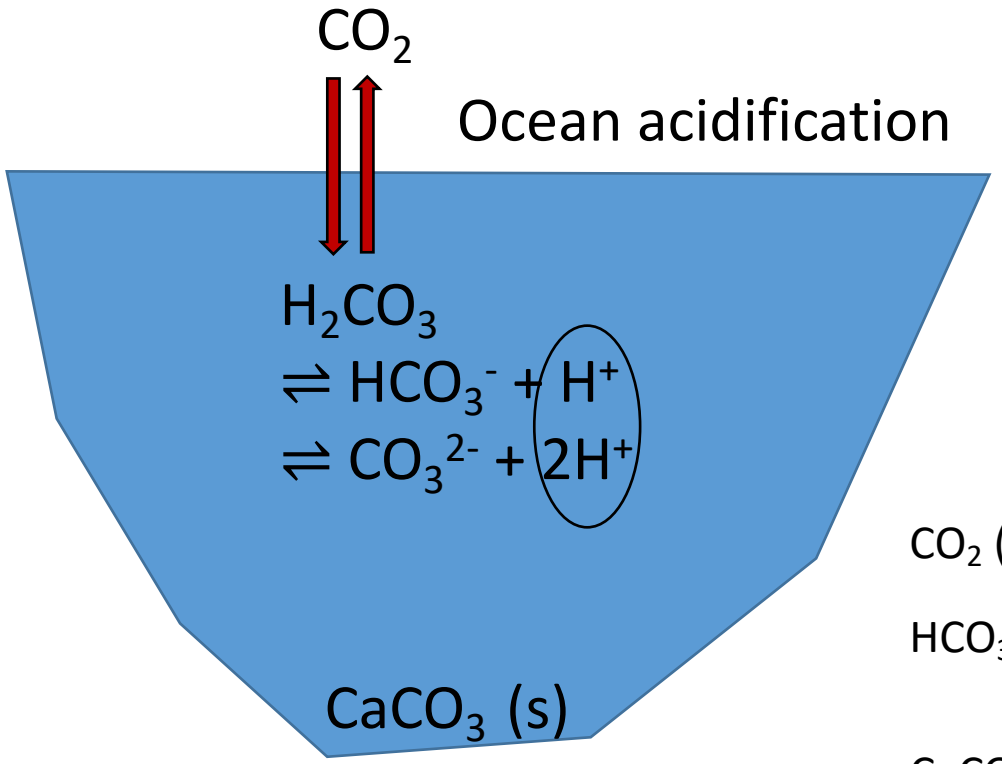
$NOx \rightarrow HNO_3 \rightarrow NO_3^- + H^+$

Chemical equilibria

- Ocean as a CO_2 sink

Dissolving $CaCO_3$ due to the ocean acidification:

$H_2O + CO_2(g) + CaCO_3(s) \rightarrow Ca^{2+} + 2HCO_3^-$



Questions:

1. What are the chemical species and their related equilibria in the ocean here?

2. Please perform the charge balance for this system.

3. If CO_2 in the air increases, what environmental problems will occur?

CO_2 (g) $\rightleftharpoons$ H_2CO_3 (aq) ↑**Pco_2** , so $[H_2CO_3]$↑ in the ocean, so **$[H^+]$↑**;

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$ ↑**$[H^+]$**, So reaction proceeds ←**left side**, so $[HCO_3^-]$↑ and **$[CO_3^{2-}]$↓**;

$CaCO_3$ (s)→$Ca^{2+} + CO_3^{2-}$ ↓**$[CO_3^{2-}]$**, so proceeds →right side, so **$CaCO_3$ (s)↓**.

Readings and practice questions

- Readings
  - Essential reading: Sec 2.2, 2.3, 2.4, 2.5, 2.6 (first 3 pages)

- Practice questions
  - Q2.1, 2.20, 2.29, 2.30, 2.35